

```
exerc.py
1  # WRITE YOUR CODE HERE
2  import os
3  import openai
4  import secret
5  from PIL import Image, ImageOps
6  from io import BytesIO
7  import requests
8  openai.api_key='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'
9
10 # Generate the base image
11 def generate_base_image(prompt):
12     response = openai.Image.create(
13         prompt=prompt,
14         n=1,
15         size="512x512"
16     )
17     return response['data'][0]['url']
18
19 cool_prompt= "Reunion of man, team, squad with katanas, ninja ,background forest"
20
21 base_image_url = generate_base_image(cool_prompt)
22 img_data = requests.get(base_image_url).content
23 with open('cool.png', 'wb') as handler:
24     handler.write(img_data)
25
26 image = Image.open('cool.png')
27 image = image.rotate(45)
28 image.save('rotated_45_cool.png')
29 image = ImageOps.invert(image)
30 image.save('final_cool.png')
31
```

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Guide

Collapse

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Coding Exercise

In this assignment, you will be performing a series of image processing tasks using DALL-E and the Python Imaging Library (PIL). You will generate a unique image, apply a rotation transformation, and perform a color inversion.

The steps are as follows:

1. Use DALL-E to generate a visually appealing image using the provided prompt. Save the image as `cool.png`.

```
cool_prompt= "Reunion of man, team, squad with katanas, ninja ,background forest"
```

2. Rotate the generated image by 45 degrees, demonstrating your understanding of basic transformations in image processing. Save the rotated image as `rotated_45_cool.png`.

3. Finally, invert the colors of the rotated image. This step will familiarize you with color space transformations. Save the final image as `final_cool.png`.
- TRY IT

Command was successfully executed.

[Click here to refresh your cool image](#)
[Click here to refresh your rotated cool image](#)
[Click here to refresh your inverted cool image](#)

Click button below when ready to submit your code.

```
# Generate the base image
def generate_base_image(prompt):
    response = openai.Image.create(
        prompt=prompt,
        n=1,
        size="512x512"
    )
    return response['data'][0]['url']

cool_prompt= "Reunion of man, team, squad with katanas, ninja ,background forest"
image_url = generate_base_image(cool_prompt)

img_data = requests.get(image_url).content
with open('cool.png', 'wb') as handler:
    handler.write(img_data)

#Saving our base image
base_image=Image.open('cool.png')
# Rotate the base image by 45 degrees
rotated_image_45 = base_image.rotate(45)

# Save the rotated image to a file
rotated_image_45.save('rotated_45_cool.png')

new_base= Image.open('rotated_45_cool.png')

# Invert the colors of the base image
inverted_image = ImageOps.invert(new_base)

# Save the inverted image to a file
inverted_image.save('final_cool.png')
```

Check It!

LAST RUN on 7/22/2023, 2:46:33 AM

Check 1 passed

Check 2 passed

Mark as Completed

Back to dashboard